

Species biology and demographic history determines species vulnerability to climate change in tropical island endemic birds

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eLife Assessment

Tropical single-island endemic bird populations are particularly vulnerable to climate change. The authors investigate genetic evidence of how such species dealt with climate changes in the past as a possible predictor for how they will respond to change in the future, which could provide an **important** example for the fields of conservation genetics and island biogeography. The authors' integration of genomics and habitat modeling is commendable, but we find that the support for their conclusions is **incomplete**: at times, the results presented appear to contradict each other, the authors do not fully account for key variables, and the limited taxonomic scope may cause problematic biases for the conclusions.

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Abstract

Climate change and associated habitat fluctuations can expedite the diversification of insular lineages, or lead to isolation and extinction. Tropical island birds are a model system to assess responses to climate change owing to their insular nature and ability to diversify rapidly. While there is some understanding of the diversification of tropical island avian clades, we are yet to understand the vulnerability of these species to climate change. Long-term species genetic diversity and demography are critical predictors of this. Therefore, we investigated the sensitivity of tropical island endemic birds to climate change and analysed how species traits determine these responses by comparing species traits with demographic histories and paleohabitat fluctuations during the Last Glacial Period (LGP). From publicly available whole genome and paleoclimatic datasets, we reconstructed tropical island endemics' past demographic histories (effective population size (N_e)) using Pairwise Sequential Markovian Coalescent (PSMC) ($n = 22$) and suitable habitat ($n = 28$) during the LGP and the Holocene. We observed that most species experienced an increase in suitable habitat from the Last Interglacial and Last Glacial Maxima. However, a concomitant increase N_e was only observed in the hyper-diverse passerine clade, attesting to their ability to rapidly diversify. Overall, diet

specialists and large-bodied species showed a decrease in N_e during the LGP. Our results indicate that species traits dictate tropical island endemics' demographic responses to climate, and plastic response to habitat availability could be a consequence of clades' abilities to rapidly occupy new niches and diversify. Further, our analyses revealed that most species entered the Holocene with low genetic diversity. Given that tropical island endemics have small geographic ranges and are groups vulnerable to climate change, special efforts are necessary to conserve them. We recommend that conservation management policies add components like historical demography and species traits while assessing extinction threats for island populations.

Introduction

Tropical islands of the Indo-Australian Archipelago (IAA), the Indo-Pacific, and the Caribbean and Atlantic have had complex geological pasts that have affected their species distribution patterns. While several Pacific islands are true oceanic islands arising *de novo* from the seafloor, many islands of the Caribbean and the IAA were once connected with each other or to the continental landmasses during periods of global sea-level fall in the Quaternary (Lohman et al., 2011; Voris, 2000). These land bridges and the additional habitat they offered during periods of sea-level fall facilitated on one hand colonisations among islands and between islands and the mainland (Andersen et al., 2015; Cros et al., 2020; Irestedt et al., 2013; Moyle et al., 2009; Ng et al., 2017; Pujolar et al., 2022) and on the other, range expansions within islands in response to the fluctuations in suitable habitat space. The habitat type and quality of these intermittently exposed land bridges varied, ranging from open savannah biomes to lowland forests (Cannon et al., 2009; Lohman et al., 2011). In the case of stratified elevational gradients, with tropical upland montane forests expanding onto lower slopes during periods of global cooling, upland montane species could disperse across otherwise persistent dual barriers of land and sea (Cannon et al., 2009).

Tropical islands' geologic pasts make them an important natural system to study species responses to changing habitat at both ecological and evolutionary timescales. Birds are an ideal clade for this because they are taxonomically well-characterised, are ecologically well-studied, and charismatic. Avian lineages have arisen on tropical islands during the Pleistocene (Andersen et al., 2015; Garg et al., 2018; Irestedt et al., 2013; Moyle et al., 2009), concomitant with complex, clade-wise colonisation and recolonisation trajectories (Filardi and Moyle, 2005; Jönsson et al., 2008, 2011b, 2014) including colonisations of the mainland (Jönsson et al., 2011a) and repeated, independent colonisation events (Cibois et al., 2011). In birds, these responses to changing habitat are governed by species traits in several continental taxa and non single-island endemics (Brüniche-Olsen et al., 2021, 2019), but responses in single-island endemics are poorly understood.

With such complex biogeographic pasts, tropical, single-island endemics represent either refugial populations, or *in-situ* diversifications as explained by taxon cycles (Ricklefs, 1970; Ricklefs and Cox, 1978, 1972). Being confined to single islands, these species are exceptionally vulnerable to ongoing anthropogenic climate change which is unprecedented and unlike past Pleistocene and early Holocene climatic changes (Crowley, 1990). This is exacerbated because we are currently experiencing a period of relatively high sea levels with reduced habitat availability for species resulting in population bottlenecks across taxa (Hewitt, 2000; Willis et al., 2004) including birds (Nadachowska-Brzyska et al., 2015; Smith et al., 2021). Information on tropical, single-island endemics' demographic responses to past climate change can inform conservation efforts, owing to the genomic signatures that predispose a species to extinction (Mays et al., 2018; Spielman et al., 2004).

Pairwise Sequential Markovian Coalescent (PSMC) is a powerful method to reconstruct the effective population size (N_e) for a species using a single diploid genome (Li and Durbin, 2011 [↗](#)) and has been used to infer demographic history across taxa (Chattopadhyay et al., 2019 [↗](#); Kim et al., 2016 [↗](#); Kozma et al., 2016 [↗](#); Murray et al., 2017 [↗](#); Nadachowska-Brzyska et al., 2015 [↗](#)). Using this along with paleo-ecological niche modelling allows us to directly correlate the demographic history of a species with its distributional range at different time points (Chattopadhyay et al., 2019 [↗](#)). We perform these analyses on a global panel of tropical single-island endemics to understand the effects of past climatic changes on species with assumed low genetic diversity owing to their status as refugial species or newly arisen lineages.

Results

Bird species panel

Our final species panel comprises 30 tropical single-island endemic bird species. Out of these, PSMC analyses were possible for 22 species and Ecological Niche Modelling (ENM) was possible for 28 species. Both PSMC and ENM analyses could be done successfully for 20 species (**Table 1** [↗](#)). These included seven Papuan species, five Philippine species, three Caribbean species, and three species from the Bismarck Archipelago. Of these 20 species, 13 species are passerines, with two from the white-eye family (Zosteropidae). Of the non-passerines, two species are parrots (family Psittacidae). All other families are represented by single species.

Paleo-habitat reconstruction

Among the 19 climatic variables used, we observed precipitation of the warmest quarter (BIO18) to contribute the most for each island endemic bird species except for the Caribbean species *Amazona guildingii*. 24 out of the 28 bird species experienced an increase in suitable habitat from the Last Interglacial (LIG) to the Last Glacial Maxima (LGM) (Supplementary table S4). 11 out of 13 Papuan species showed an increase in habitat availability from the LIG to the LGM, with *Cicinnurus regius* and *Pseudorectus ferrugineus* as exceptions. All Philippine species showed an increase in habitat from the LIG to the LGM as well (**Figure 1** [↗](#), Supplementary information S2). Two out of the three Caribbean species experienced a decline in suitable habitat during this period. Of the remaining species, all showed an increase in available habitat. Upland Papuan species remained confined to upland montane regions at all the time points we reconstructed available habitat, with *Rhagologus leucostigma* as a notable exception (Supplementary information S2). Papuan lowland species remained largely confined to regions which were not newly-exposed landbridges at all times as well. From the LGM to the present day, habitat decreased for 19 out of the 28 bird species, and two species experienced no change (Supplementary table S3). For the present study, we primarily concentrate on habitat fluctuation between the LIG and the LGM as this is the period for which we have comparative evidence of fluctuations of both habitat as well as genetic diversity.

Strong Quaternary fluctuations in genetic diversity

PSMC analyses could be successfully done for 22 species (**Table 1** [↗](#)). Results indicate large genetic diversity variations in almost all bird species, corresponding to the Quaternary climate shifts (**Figure 2** [↗](#)). The demographic history of these species reconstructed using Pairwise Sequentially Markovian Coalescent (PSMC) analyses extends back over a million years (Supplementary information S1). Reconstructed N_e values were generally concordant across the three PSMC settings used, except for five species where N_e values had large differences across PSMC settings for N_e values at the LIG and LGM. However, in these species as well, trends of N_e increase or decrease from the LIG to LGM were robust across all three PSMC models considered. *Centropus unirufus*, *Dicaeum eximium*, *Irena cyanogastra*, *Sterrhoptilus dennistouni*, *Zosterops hypoxanthus*

Species	PSMC	ENM	Endemic Region	Passerine	Family
<i>Actenoides hombroni</i>	y	y	Philippines	n	Alcedinidae
<i>Aleadryas rufinucha</i>	y	y	Papua New Guinea	y	Oreoicidae
<i>Alectura lathami</i>	y	y	Australia	n	Megapodiidae
<i>Amazona guildingii</i>	y	y	St Vincent	n	Psittacidae
<i>Amazona vittata</i>	y	y	Puerto Rico	n	Psittacidae
<i>Amblyornis subalaris</i>	y	n	Papua New Guinea	y	Ptilonorhynchid ae
<i>Cacatua alba</i>	y	n	Indonesia	n	Cacatuidae
<i>Centropus unirufus</i>	y	y	Philippines	n	Cuculidae
<i>Chaetorhynchus papuensis</i>	n	y	Papua New Guinea	y	Rhipiduridae
<i>Cicinnurus regius</i>	y	y	Papua New Guinea	y	Paradisaeidae
<i>Cnemophilus loriae</i>	y	y	Papua New	y	Cnemophilidae

Table 1

Details of the taxa included in this study.

PSMC = Pairwise Sequential Markovian Coalescent, ENM= Ecological Niche Modelling, y = yes, n = no.

			Guinea		
<i>Dicaeum eximium</i>	y	y	Bismarck Archipelago	y	Dicaeidae
<i>Diphyllodes magnificus</i>	y	y	Papua New Guinea	y	Paradisaeidae
<i>Eulacestoma nigropectus</i>	n	y	Papua New Guinea	y	Eulacestomatid ae
<i>Ifrita kowaldii</i>	n	y	Papua New Guinea	y	Ifritidae
<i>Irena cyanogastra</i>	y	y	Philippines	y	Irenidae
<i>Machaerirhynchus nigripectus</i>	y	y	Papua New Guinea	y	Machaerirhync hidae
<i>Myiagra hebetior</i>	y	y	Bismarck Archipelago	y	Monarchidae
<i>Oreocharis arfaki</i>	n	y	Papua New Guinea	y	Paramythiidae
<i>Pachycephala philippinensis</i>	y	y	Philippines	y	Pachycephalida e
<i>Paradisaea raggiana</i>	y	y	Papua New Guinea	y	Paradisaeidae
<i>Paradisaea rubra</i>	n	y	Indonesia	y	Paradisaeidae

Table 1 (continued)

<i>Parotia lawesii</i>	n	y	Papua New Guinea	y	Paradisaeidae
<i>Pseudorectes ferrugineus</i>	n	y	Papua New Guinea	y	Pachycephalidae
<i>Ptilorrhoa leucosticta</i>	n	y	Papua New Guinea	y	Cinclosomatidae
<i>Rhagologus leucostigma</i>	y	y	Papua New Guinea	y	Rhagologidae
<i>Rhynochetos jubatus</i>	y	y	New Caledonia	n	Rhynochetidae
<i>Sterrhoptilus dennistouni</i>	y	y	Philippines	y	Zosteropidae
<i>Todus mexicanus</i>	y	y	Puerto Rico	n	Todidae
<i>Zosterops hypoxanthus</i>	y	y	Bismarck Archipelago	y	Zosteropidae

Table 1 (continued)

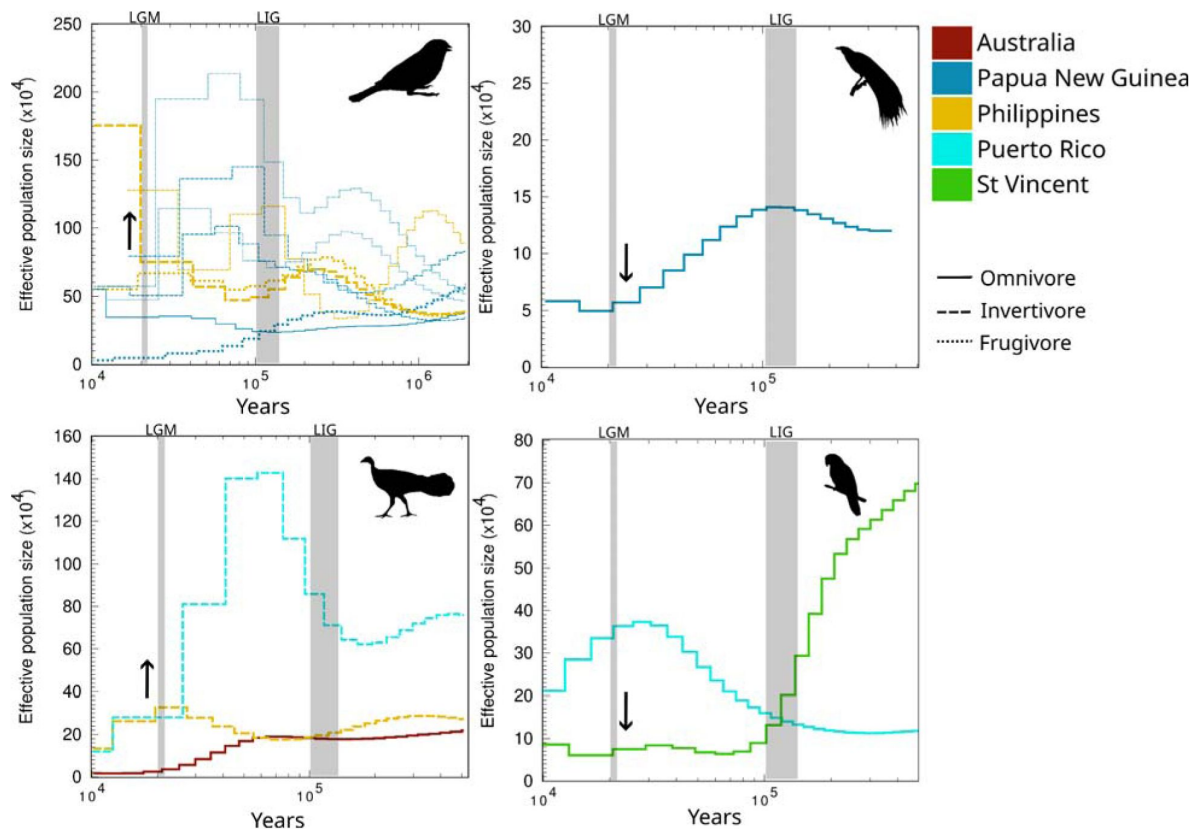


Figure 1

Pairwise Sequential Markovian Coalescent (PSMC) plots using the settings -p "2 + 2 + 30 * 2 + 4 + 6 + 10" displaying reconstructed effective population size values with time for passerines (above) and non-passerines (below) based on whether habitat availability increased (left) or decreased (right) during the Last Glacial Period (LGP).

Colours indicate the archipelago/island the bird belongs to, and the line style indicates the dietary habit of the bird species. Bold lines indicate large (> 50 g body mass) bird species. The grey bands indicate the approximate durations of the last interglacial (LIG) and the last glacial maxima (LGM). Black arrows indicate if the habitat availability increased or decreased during the LGM. A mutation rate of 1.4×10^{-9} years/site and a generation time of 2 years for passerines, and a mutation rate of 1.91×10^{-9} years/site and a generation time of 1 year for non-passerines were used to generate plots. *Zosterops hypoxanthus* is not displayed because its Ne values far exceed those of the other species.

were the six species which had large differences (Supplementary information S1). We found no significant difference between the different sets of PSMC settings used (Kruskal-Wallis test, $\chi^2 = 0.217$, $DF = 2$, $p = 0.897$).

The 14 passerine species we analysed showed overall lower N_e values in the LGM as compared to the LIG (**Figure 1**). We obtained similar results for non-passerines as well, with *Rhynochetos jubatus* as a notable exception using the $-p$ “4 + 30 * 2 + 4 + 6 + 10” setting, displaying a large peak followed by a crash in N_e (Supplementary information S1). This is likely a PSMC artefact, because plots using other parameter settings did not show peak (Supplementary information S1). Along with overall lower values, we also see a smaller range of N_e values in the LGM as compared to the LIG (**Figure 1**).

Correlation between historical fluctuations in N_e and distribution

Habitat change was poorly associated with change in N_e for the 20 species for which both PSMC and ENM analyses were possible (Cramer’s $V = 0.15$). However, passerine species only showed a strong association (Cramer’s $V = 0.96$), while non-passerines showed a weak negative association (Cramer’s $V = -0.15$).

Bayesian multivariate regression models revealed an overall fluctuation in N_e was associated with species biology and habitat change during the Last Glacial Period (LGP) even after controlling for the geographical island group (**Figure 3**). Most confidence intervals and beta coefficients did not overlap with zero (**Figure 3**) suggesting significant relationships. We found a positive association between change in habitat area (LIG to LGM) and N_e ($\beta = 9.45$, 95% CI: [1.95, 21.1]). This means that species with increasing habitat area in the LGP also showed an increase in N_e . Change in N_e and body mass showed a significant negative association ($\beta = -8.63$, 95% CI: [16.27, -2.61]). That is, large-bodied species showed decreases in N_e during the LGP. Diet specialists like frugivores ($\beta = -11.89$, 95% CI: [-22.65, -2.99]) and invertivores ($\beta = -11.7$, 95% CI: [-22.66, -2.94]) also showed significant negative associations with population change. However, omnivores ($\beta = -1.39$, 95% CI: [-13.38, 11.5]) did not show any association with population change. Whether or not a species was a passerine was an important predictor of N_e only in combination with the change in habitat from LIG to LGM ($\beta = 7.97$, 95% CI: [1.82, 16.91]), suggesting that passerines respond positively to habitat change, as suggested by Cramer’s V as well. The interaction between habitat change and body mass ($\beta = 10.05$, 95% CI: [-0.3, 24.41]) suggests that there is no impact of habitat change in larger species. Finally, the random intercept for Country (sd (Intercept)) showed a marginal positive influence ($\beta = 0.85$, 95% CI: [0.04, 2.24]) (**Figure 3**, Supplementary table S6).

Discussion

Island birds are vulnerable to the effects of climate change

Our results on the fluctuations in paleohabitat and genetic diversity (N_e) during the LGP highlight the vulnerability of island birds to climatic fluctuations. Climate fluctuations change habitat availability and, in turn, genetic diversity, making these species particularly sensitive to a changing environment. In response to landmass expansion, several species in our panel across islands showed an increase in their genetic diversity during the LGP (**Figure 1**, Supplementary information S1), although the overall relationship is weak (Cramer’s $V = 0.15$). This could be because large amounts of intermittently exposed landbridge habitat remain uninhabited by the species in our panel (Supplementary figure S2). For example, we know that large amounts of exposed landbridge habitat available to our panel’s Papuan species range from lowland evergreen to seasonal rainforest, with an expansion of upland habitats of evergreen to seasonal rainforest. Transitional hill forest formed an ecotone between these two (Cannon et al., 2009). However, most of our species inhabit dense forest (Supplementary table S7), which is unlikely to comprise newly exposed landbridge habitat.

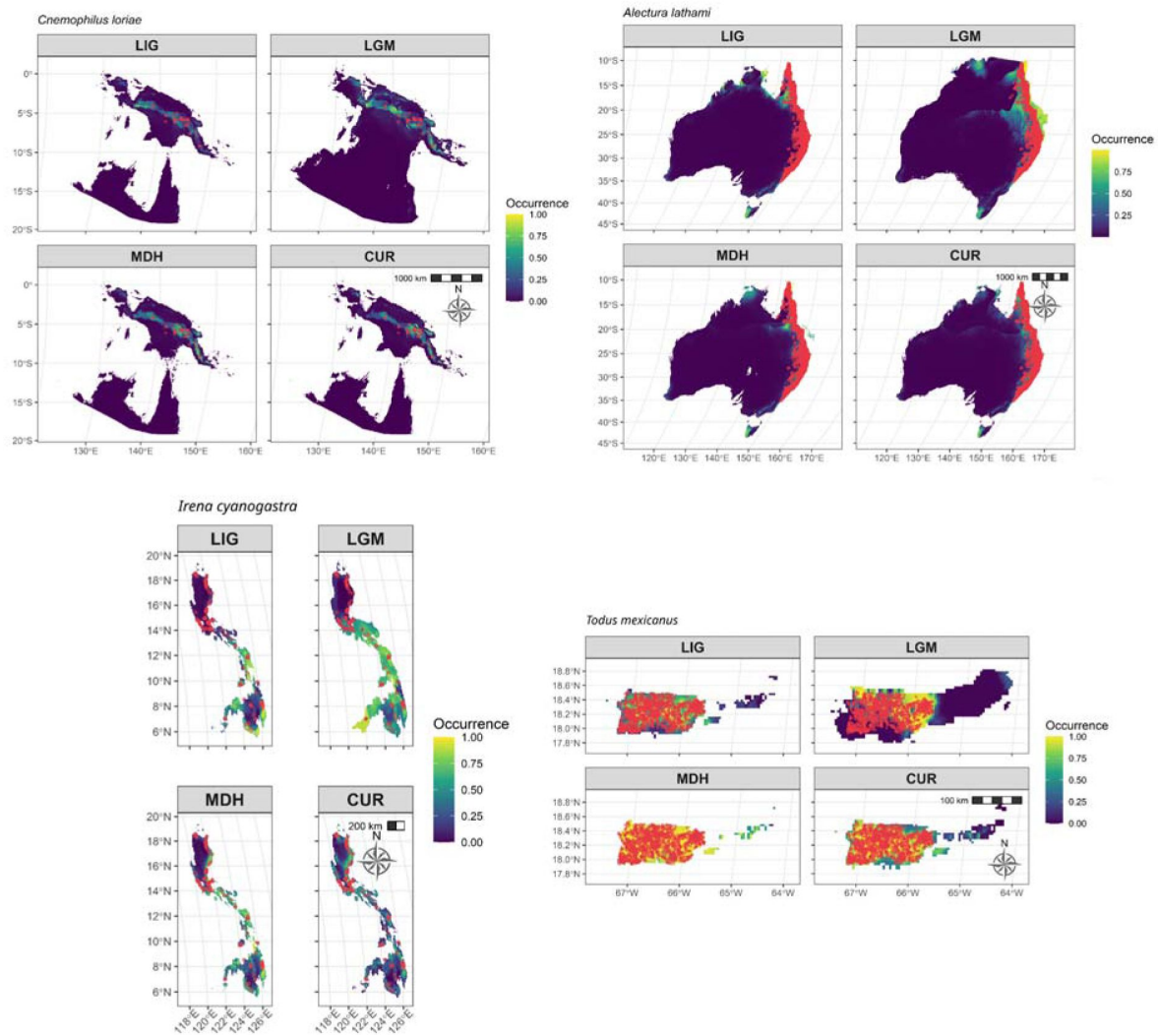


Figure 2

Example of Ecological Niche Modelling plots for species from Papua (*Cnemophilus loriae*), Australia (*Alectura lathami*), the Philippines (*Irena cyanogastra*), and the Caribbean (Puerto Rico, *Todus mexicanus*).

LIG = Last Interglacial. LGM = Last Glacial Maximum. MDH = Mid Holocene. CUR = Current. The continuous heatmap represents the probability of occurrence of the species and red points are known occurrences from GBIF. For all the plots see figure S2.

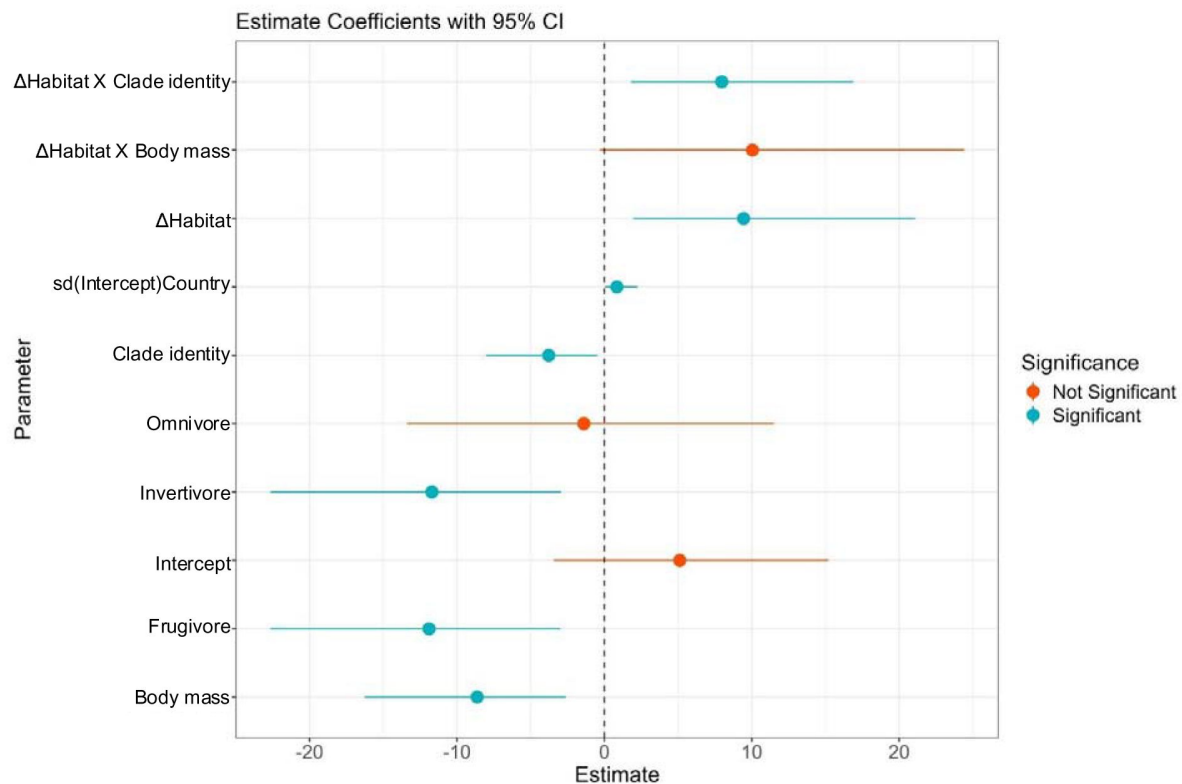


Figure 3

Results of the best Bayesian multivariate regression model performed.

The response variable represents the changes in effective population size (increased (=1) or decreased (=0)) during the last glacial period. Circles represent mean values and lines represent 95% confidence intervals. Model parameters and coefficients are provided in table S6. ΔHabitat = the change in suitable habitat from the Last Interglacial to the Last Glacial Maximum.

Our results also reveal that both passerine and non-passerine island endemics have entered the Holocene with low genetic diversity (**Figure 1**). A loss of genetic diversity predisposes species to extinction (Mays et al., 2018; Spielman et al., 2004) with a potentially causal relationship for birds (Evans and Sheldon, 2008). The current climate crisis has greatly imperilled biodiversity with the documented loss of many vertebrate species (Steadman and Martin, 2003; Tan et al., 2023; Willis et al., 2004). Fossil evidence has revealed the extinction of several species in the Caribbean since the LGM (Morgan and Woods, 1986; Orihuela et al., 2020; Steadman et al., 1984), and many of the species in our panel are Southeast Asian where rapid habitat loss is ongoing (Sodhi et al., 2004). Climate warming, sea level rise and changes in vegetation are all associated with this loss, and flightless birds and endemics are particularly prone to extinction (Fromm and Meiri, 2021).

Habitat loss, rise in sea level and warming temperatures can rapidly accelerate species extinction, particularly in the tropics (Şekercioğlu et al., 2012). Effective population size, and in extension genetic diversity are associated with survival of a species and its extinction risks (Frankham, 2005). Coalescent effective population size as used in PSMC is an important predictor of species vulnerability to extinction (Brüniche-Olsen et al., 2021; Wilder et al., 2023). Thus, an evolution-informed understanding of species vulnerability to climate change and their associations with paleohabitats can be an important tool to predict species vulnerability to the current climate crisis and future extinction risk (Brüniche-Olsen et al., 2021; Chattopadhyay et al., 2019; Gabrielli et al., 2024; Germain et al., 2023).

A strong, passerine association might be driven by their rapid diversification

Passerines are a hyperdiverse clade representing over 60% of extant avian diversity. The most recent, fossil-calibrated passerine phylogeny (Oliveros et al., 2019) shows that passerines began diversifying in the Middle (47 mya) to Late Eocene (38–39 mya) with increasing diversification rates suggested as we move towards the present. Crown passerines originated in the Australo-Pacific region (Oliveros et al., 2019), and this places the many Papuan and Australian species in our panel close to the passerine diversification centre which is known to have rapid diversification rates (McCullough et al., 2022). Because bird lineages are known to have arisen on tropical islands in the Pleistocene (Andersen et al., 2015; Irestedt et al., 2013; Moyle et al., 2009), it is likely that the Southeast Asian species in our panel represent newly arisen lineages over refugia, possibly belonging to currently rapidly-diversifying clades. This is supported by the fact that several of these species belong to small, often oligotypic families such as Oreocidae (*Aleadryas rufinucha*), Ptilonorhynchidae (*Amblyornis subalaris*), Cnemophilidae (*Cnemophilus loriae*), Eulacestomatidae (*Eulacestoma nigropectus*), Ifritidae (*Ifrita kowaldi*), Machaerirhynchidae (*Machaerirhynchus nigripectus*), and Rhagologidae (*Rhagologus leucostigma*). Many of these species belong to poorly studied tropical genera for which species level divergence times are unavailable. The strong passerine association between change in N_e and available habitat area (Cramer's $V = 0.96$; **figure 3**, $\beta = 9.45$) reveals that passerines are strongly plastic to the environment, most possibly as a consequence of their ability to rapidly diversify.

Non-passerines representing a paraphyletic outgroup do not show this trend (Cramer's $V = -0.15$). The seven non-passerine species for which both ENM and PSMC analyses were possible belong to six different families (**Table 1**), representing a diverse sample across the avian phylogeny and precluding non-passerine phylogenetic insight. Notably, we confirm (Cramer's $V = 0.15$) that habitat area is positively correlated with genomic diversity in a clade non-specific manner for birds in general (Brüniche-Olsen et al., 2021, 2019).

Species traits influenced historic fluctuations in Ne during the LGP

The interplay of species traits and habitat availability determine how Ne values fluctuate with time in non-endemic birds (Brüniche-Olsen et al., 2021 [↗](#)), and our results confirm this for tropical island endemics as well. Habitat change in the LGP was positively associated with Ne fluctuations (Figure 3 [↗](#), $\beta = 9.45$), that is, species which showed an increase in habitat in the LGP also showed a concurrent increase in Ne. While diet generalists like omnivores showed no significant association with Ne fluctuations, specialists like frugivores (Figure 3 [↗](#), $\beta = -11.89$) and invertivores (Figure 3 [↗](#), $\beta = -11.7$) showed significant negative association with fluctuation in Ne, implying that specialist species tended to show decreases in Ne in the LGP. Previous work has shown that large-bodied species have lower values of standing genetic diversity at a given time point (Brüniche-Olsen et al., 2021 [↗](#), 2019 [↗](#); Eo et al., 2011 [↗](#)). We find large-bodied species to show a negative association with Ne fluctuations (Figure 3 [↗](#), $\beta = -8.63$) as well. This implies that Ne for large-bodied species have decreased in the LGP. Traits like body size and diet determine extinction risk (Ripple et al., 2017 [↗](#); Willis et al., 2004 [↗](#)), and our results add to the understanding of how traits modulate Ne changes through time.

Assessing the relative contribution of species traits and habitat availability in determining Ne requires their quantitative measurements. Our methods generated values for available habitat at various time points (Supplementary table S3). However, Ne values can so far only be reconstructed using coalescent methods like PSMC. These values depend upon the parameter settings used. While we find these to be generally concordant across settings (Kruskal-Wallis, $p = 0.897$, Supplementary table S1), the estimates vary. We therefore chose to cautiously measure only the direction of change (increase or decrease) and not use numerical values.

Caveats of PSMC analyses

Coalescent methods for inferring demographic history like PSMC can trace a local population's demographic history rather than the entire species history (Gattepaille et al., 2013 [↗](#); Heller et al., 2013 [↗](#); Ptak and Przeworski, 2002 [↗](#)). Population structure is thus a confounding factor. However, this is unlikely to be a problem given that all our species are single-island endemics. Large peaks followed by apparent collapses in Ne have also recently been shown to be an artefact of PSMC analyses (Hilgers et al., 2024 [↗](#)). We try three different parameter combinations, and look for consensus amongst them in order to try and mitigate this. Because reconstructed Ne values are inferred based on these parameter settings, we do not use them directly for analyses, instead looking at the direction of population size change (increase/decrease/no change). Lastly, we emphasise that Ne is a scaled value of the real population size (Wang et al., 2016 [↗](#)), and is measured to be up to one-tenth of the real population size in wild populations (Frankham, 1995 [↗](#)). We emphasise caution in interpreting Ne results as a proxy for real population size.

Methods

Species selection

We queried Avibase (Lepage et al., 2014 [↗](#)) for all species endemic to tropical islands and selected all species for which assembled whole genome sequences were available on GenBank using short-read data from the Illumina platform. We only selected species that had not been analysed using similar methods already. We excluded genomes assembled from museum specimens. This resulted in a panel of 30 species (Table 1 [↗](#)).

Pairwise Sequential Markovian Coalescent (PSMC) analyses

For each assembled diploid genome, we identified the contigs corresponding to the sex chromosomes using Satsuma ver. 2.0 (Grabherr et al., 2010) by aligning contigs to a *Gallus gallus* (chicken; BioSample: SAMN02981218) genome as a reference. All contigs mapping to the sex chromosomes were removed. Next, we obtained all available Illumina short reads for each species from the Sequence Read Archive (SRA) database of the NCBI, and checked them for errors using FastQC (Andrews, 2010). We used Trimmomatic (Bolger et al., 2014) for preprocessing and trimming the reads. The cleaned reads were aligned to the autosomes using BWA-MEM ver. 0.7.17 (Li, 2013). Next, we used SAMtools ver. 1.10 (Danecek et al., 2021) to merge all the resulting files to generate a single bam file for each species. We further sorted and removed duplicates using SAMtools and estimated the depth of coverage for each species. We excluded species with a depth of coverage < 18X. This is a sufficient value of coverage for PSMC analyses (Li and Durbin, 2011). Next, we implemented the SAMtools mpileup-bcftools pipeline to identify SNPs. The minimum and maximum depths for calling SNPs was set to one-third and double the mean depth respectively following Nadachowska-Brzyska et al., (Nadachowska-Brzyska et al., 2015).

We used three sets of parameters for PSMC analyses and looked for consensus trends amongst them because PSMC analyses can result in spurious peaks of effective population size (Hilgers et al., 2024). We used the following parameter sets for PSMC analyses: $-t5 -b -r1 -p "4 + 30 * 2 + 4 + 6 + 10"$, $-t5 -b -r1 -p "2 + 2 + 30 * 2 + 4 + 6 + 10"$, and $-t5 -b -r1 -p "1 + 1 + 1 + 1 + 30 * 2 + 4 + 6 + 10"$. We performed 30 iterations for parameter optimisation and ran 100 bootstrap replicates to judge the uncertainty in our estimates. Bootstrapping was done by randomly sampling chromosome segments with replacement and running PSMC on them. To plot the results from the PSMC analyses, we used previously estimated values of mutation rates and generation times for passerines (1.4×10^{-9} years/site and 2 years respectively; (Ellegren et al., 2012; Nadachowska-Brzyska et al., 2016) and non-passerines (1.91×10^{-9} years/site and 1 year respectively) (Nam et al., 2010) (Supplementary material S1, supplementary table S1).

Species and climate records

We performed reconstructions of paleohabitats through ecological niche models and reconstructed species distributions from four time periods: the Last Interglacial (LIG, approx. 110,000–130,000 years ago), the Last Glacial Maximum (LGM, approx. 20,000 years ago), the Mid-Holocene (MDH, approx. 6000 years ago) and Current (CUR, present day).

We used R (ver. 4.2.1; R Core Team, 2021) for all paleo-habitat modelling analyses. We accessed the location records of 20 bird species from the Global Biodiversity Information Forum (GBIF) in September, 2023 using the “rbgif” package (ver. 3.7.9) (Chamberlain and Boettiger, 2017) in R. We discarded duplicates and retained only human observed records using the “tidyverse” (v-2.0.0) package (Wickham and RStudio, 2023) followed by the ‘CoordinateCleaner’ package (v-3.0.1) (Zizka et al., 2023). We used the “spThin” package (Aiello-Lammens et al., 2014) to account for spatial autocorrelation and finally discarded spurious distribution records like records overlaying water bodies, buildings, roadways, and railways using QGIS (v-3.34+; <https://www.qgis.org/>).

We used 19 climate variables (labelled BIO1 – BIO19; Supplementary table S2) for the four different time periods. We downloaded CHELSA (<https://chelsa-climate.org/>; Karger et al., 2021) climate predictors at a resolution of 2.5 arc-minutes (~5 km) for all regions except for *Amazona guildingii* from Saint Vincent where we used a resolution of 30 arc-seconds (~1km). For the latter, we used WORLDCLIM (<https://www.worldclim.org/>; Fick and Hijmans, 2017) for data for the LIG and MDH while CHELSA was used for the LGM and CUR.

We followed [Chattopadhyay et al., \(2019\)](#) and used a global dataset approach to account for idiosyncratic biases due to smaller datasets to extract the location points of each endemic-island group against each climatic variable. We tested for multicollinearity using variance inflation factor (VIF) using the ‘usdm’ R package ([Naimi, 2023](#)) and considered variables for further analyses if their VIF value was ≤ 5 .

Habitat Suitability Modelling and Area Calculation

We used R to reconstruct paleo-climatic suitable habitat for 28 endemic island birds through ecological niche models for the four time periods considered. For this, we accessed the Global Biodiversity Information Forum (GBIF) to extract our species’ occurrence records. We could not perform ENM analyses for two species due to a paucity in the number of GBIF occurrence points for them ([Table 1](#)).

Data partitioning and model evaluation

For each species, we used available occurrence records from the present day to generate pseudo-absence data points. We generated 500-10,000 pseudo-absence/background points for each species depending upon the area of the island (Supplementary table S3), except for *Amazona guildingii* endemic to Saint Vincent where we generated 40 points. Saint Vincent’s small land area could not accommodate more points than this. We used a subset of bioclimatic variables to generate a weighted average ensemble species distribution model (eSDM) for each island group. eSDMs were implemented in the R package “sdm” (v-1.2.37; [Naimi and Araujo, 2016](#)) applying the ‘MaxEnt’, ‘GLM’, and ‘BRT’ algorithms. eSDMs account for the limitations of different models by generating a weighted average of multiple models ([Araújo and New, 2007](#); [Dormann et al., 2018](#); [Naimi and Araujo, 2016](#)). We used k-fold cross-validation (CV) with replication for each method for training and test datasets across each endemic island group (Supplementary table S3). Model accuracy was measured using the Area Under the Curve (AUC) and True Skill Statistics (TSS) metrics. We first performed the above analyses for the present-day (CUR) distribution using the weighted-average of AUC and TSS. AUC and TSS values greater than 0.9 and 0.75 respectively have been shown to be indicators of superior model performance ([Ahmad et al., 2019](#)) and we chose a similar threshold ($AUC \geq 0.9$ & $TSS \geq 0.8$) to get the best ensemble model for most of the species except for 10 species for which data quality was poor (Supplementary table S3). Models for the other time points: the LIG, LGM, and MDH, were generated using the eSDM model generated based on CUR data.

Suitable area analysis

All spatial analyses were carried out using the R package ‘terra’ (ver. 1.7.13) ([Hijmans et al., 2024](#)). Equal area projections were used to calculate the absolute suitable area of each species across the four time-periods. To account for a spherical Earth and the limitations of landmass depiction of the original eSDM raster (World Geodetic System 1984) we reprojected it into a pseudocylindrical projection (Eckert IV). Further, we used the average quantile threshold pixels to measure the availability of absolute suitable areas for the LIG, LGM, MDH, and CUR periods (Supplementary table S4).

Statistical Analyses

We checked for associations between N_e and habitat change during the Last Glacial Period (from LIG to LGM) (LGP) using Cramer’s V implemented in the “polycor” package in R ([Fox, 2022](#)). Cramver’s V varies between -1 to 1 , with values closer to 0 suggesting no-correlation. This was done for the entire dataset, and then for passerines and non-passerines separately.

We further explored the effects of species biology, habitat, and phylogenetic constraint on the fluctuation in N_e during the LGP using Bayesian Multilevel Models (MLMs), using the “brms” package in R ([Bürkner, 2018](#)). Our fixed effect predictors were habitat change from the LIG to LGM, diet (invertivore, frugivore, or omnivore), body mass, and clade identity (passerine and non-

passerine). The latter allows us to check if belonging to the rapidly diversifying passerine clade results in a signal. We also included country (the island/archipelago a species is endemic to) as a random effect variable for the analysis to account for country-specific effects. For the response variable i.e., the change in N_e , a Bernoulli distribution with a logit link because it is a binary response variable.

We ran a series of models ($n=12$) to find the best-fit model (brms11, Supplementary table S6) based on leave-one-out cross-validation. We generated a total of 16,000 posterior samples by using the No-U-Turn Sampler (Hamiltonian Monte Carlo) algorithm with 4 chains with 5000 iterations each (with 1000 warm-up iterations; supplementary information S3). Further, we estimated the posterior parameters with 95% confidence intervals to find negative or positive associations between the N_e and its predictors. In addition, we assessed the posterior convergence of the sample through R -hat statistics ($R^{\wedge} = 1$) where values close to 1 suggest an ideal convergence. We also reported the effective sample size for bulk and tail distributions (Supplementary table S5).

Data availability

The raw data used for this study is available on NCBI and details of the accession ID are provided in Table S1. All codes used in this study are provided on the Zenodo URL <https://doi.org/10.5281/zenodo.14603966> .

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Additional files

Supplementary figures 

Supplementary tables 

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Reviewer #1 (Public review):

Summary:

The authors combine PSMC and habitat modeling to try to connect habitat change during the Last Glacial Period to changes in N_e .

Strengths:

Observing how tropical single-island endemic bird species responded to habitat change in the past may help inform conservation interventions for these particularly vulnerable species. The combination of genomics and habitat modeling is a good idea - this sort of interdisciplinary thinking is what is needed to tackle these complex questions. Additionally, the use of PSMC makes it possible to perform this analysis on poorly-studied species with only a single genome available.

Room for Improvement:

Why coalescent N_e is a better predictor of extinction risk than current genomic diversity, or current N_e , isn't explicitly explained. PSMC in particular has many caveats, and some are not acknowledged or adequately addressed by the authors. For example, the authors note that population structure is a confounding factor with PSMC, but that it is not a problem in this instance. They do not provide compelling evidence for why this would be the case, they simply state that the species studied are all single-island endemics. However, single-island endemic species are not necessarily panmictic; this is even less likely to be true for species studied here that inhabit a large geographic area (ie, Australian species). Differing PSMC parameters may also impact results: the differences between passerines and non-passerines were one of their main results, but they do not provide any analysis to show that this difference was not driven by the different mutation rates used for the two groups.

Parameters for many steps are not described, and choices that are described (such as the PSMC parameters) are not always fully explained. It is unclear why all data was mapped to the autosomes rather than removing reads that map to the sex chromosomes first. Using all the data, the reads belonging to the sex chromosomes could potentially map to other areas of the genome. It does not seem like a mapping quality filter was used, so these potential spurious alignments would not have been removed prior to analysis.

There are points where the results are described in ways that appear to potentially differ from the supplementary figures. The authors state that even for species where PSMC results differed between models, "trends of N_e increase or decrease from the LIG to LGM were robust across all three PSMC models considered." The figures in the supplement for *Pachycephala philippinensis*, *Rhynchoceros jubatus*, and *Zosterops hypoxanthus* appear to potentially contradict this statement, but it is difficult to tell, as the time period observed is not clearly marked on the graphs. How this robustness of trends was determined is not explained, leaving the precision of the analysis unclear.

Table 1 also includes some information that contradicts what is in the Supplementary Tables, leading to a lack of clarity. *Centropus unicolor*, *Chaetorhynchus papuensis*, and *Cnemophilus loriae* are not included in Supplementary Table 4. Table 1 says *Eulacestoma nigropectus*, *Paradisaea rubra*, and *Parotia lawesii* did not undergo PSMC analysis, but Supplementary Table 4 says PSMC and modeling trends matched for these species. Table 1 says *Rhagologus leucostigma* underwent both PSMC and climate modeling, but Supplementary Table 4 says "NA" as if it was missing one of these analyses.

Additionally, some of the results appear to contradict each other. For example, they show that there is no impact of habitat change in larger-bodied species, but also that larger-bodied species saw a decrease in N_e during the LGM. In another example, they state that when a species saw an increase in habitat during the LGM, they also had an increase in N_e . However, they also state that this was not the case for non-passerines.

Ecosystems are highly complex; there may also be other variables influencing past demographic change other than those explored here. Results should be interpreted with caution.

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Reviewer #2 (Public review):

Summary and strengths:

In this manuscript, Karjee and colleagues used coalescent-based effective population size reconstruction (PSMC) from single genomes to understand past population trends in island birds and related this to life history traits and glacial patterns. This concept is fairly new, as there are still relatively few multiple PSMC synthesis studies. I also thought that the focus on island endemics was unique and adds value to this paper. I enjoyed seeing a paper focused on South East Asia and think that this could help contribute to our knowledge of the important biodiversity within this region.

Major weaknesses:

My biggest concern with this paper is that the analyses are limited to 20-30 species, and significant taxonomic bias is present (there are multiple species of passerine but only 1-2 representatives of other groups). While this is not an issue alone, many of the life history traits or geographical traits are conflated with phylogenetic diversity (e.g., there are no large-bodied passerines). Thus, it is my opinion that the impact of these drivers of past population size is conflated and cannot be disentangled with the current data. The authors themselves state that the core hypothesis surrounding N_e and habitat availability is not supported by their entire dataset (only seen in Passerines). This was not clear enough in the abstract, and conclusions cannot be drawn here as the impact of taxonomy cannot be separated from data richness, traits, etc. The PSMC analysis was done according to the most recent recommendations, and this part of the manuscript is fairly robust. However, in several places, it is incorrectly stated that the PSMC measures or can infer genetic diversity; PSMC

only infers past effective population size. It cannot measure genetic diversity in the past. I cannot review the habitat reconstruction modelling as I am a conservation genomics specialist.

Appraisal:

I am not convinced about the findings within the paper. I do not think that the results are sufficiently supported at this time, largely due to the conflation of taxonomy with other variables. As this type of comparison is new, I do think that there is a chance for reasonable impact on the field of genomics and island biogeography if the manuscript's constraints are addressed. I do not see scope for impact on conservation at this time and find the conclusions in the abstract regarding conservation relevance to be unfounded.

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Author response:

We thank the editors and the reviewers for their positive comments regarding our manuscript and the methodological approach we have taken to understand the historical demographic response of endemic island birds to climate change. We acknowledge the issues of uneven sample sizes and plan to include additional species of island endemic birds for which genomic data is now available. As requested by reviewer 1, we will also address the issues related to the PSMC analysis in the revised version of the manuscript.

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